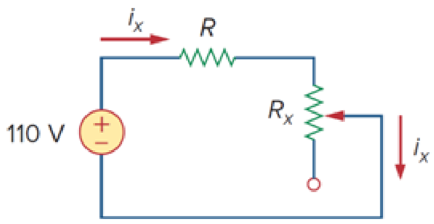
Chapter 2, Problem 64P

Problem

The potentiometer (adjustable resistor) *Rx* in Fig. 2.126 is to be designed to adjust current *ix* from 10 mA to 1 A. Calculate the values of *R* and *Rx* to achieve this.

Figure 2.126:



Step-by-step solution

Step 1 of 2

Refer to Figure 2.126 in the textbook.

The design constraint of the potentiometer is that the current i_x should be in the range from 10 mA to 1 A.

For the current i_x to be maximum (1 A), the potentiometer should be at its minimum resistance, that is, zero.

Now the resistance in the circuit is R.

The current through the circuit, i_x is equal to the current through the resistance, R. Thus,

$$\frac{110}{R} = i_x$$
$$\frac{110}{R} = 1$$
$$R = \frac{110}{1}$$
$$R = 110\ \Omega$$

Therefore, the value of resistance *R* is $\boxed{110\ \Omega}$.

Step 2 of 2

For the current i_x to be minimum (10 mA), the potentiometer should be at its maximum resistance, that is, R_x.

Now, the resistance in the circuit is $R+R_x$.

The current through the circuit, i_x is equal to the current through the resistance, $R+R_x$.

Thus,

$$\frac{110}{R+R_x}=i_x$$
$$\frac{110}{R+R_x}=10\times10^{-3}$$

Substitute $110\ \Omega$ for R.

$$\frac{110}{110+R_x}=0.01$$
$$110=1.1+0.01R_x$$
$$0.01R_x=108.9$$
$$R_x=10.89\ \text{k}\Omega$$

Therefore, the value of resistance R_x is $\boxed{10.89\ \text{k}\Omega}$.